No. of Functional Features Included in the Solution

Date	8 July 2026
Team ID	
Project Name	**OptiCrop: Smart Agricultural Production Optimization Engine**
Maximum Marks	5 Marks

Functional Features Overview:

S.No	Feature Name	Feature Description	Module / Compon en t	Status (Done / In Progress / Pending)	Marks Contributi on
1	Crop Recommendation	Predicts the most suitable crop using soil and environmental parameters.	Machine Learning Model	Done	High
2	Soil Parameter Input	Allows users to enter Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall values.	Frontend	Done	High
3	Input Validation	Validates user inputs before processing predictions.	Flask Backend	Done	Medium
4	Data Preprocessing	Cleans and formats input data before sending it to the ML model.	Backend Logic	Done	High
5	Machine Learning Prediction	Loads the trained .pkl model and generates crop recommendations.	Prediction Module	Done	High
6	Responsive User Interface	Displays input forms and prediction results in a user-friendly interface.	HTML, CSS, JavaScript	Done	Medium
7	Local/Online Deployment	Deploys the application for users through the web.	Flask / Vercel	Done	Medium
8	Future Enhancements	Weather forecasting, fertilizer recommendation, and multilingual support.	Future Scope	Pending	Low

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Soil parameter input, prediction result display
2	Backend / Logic	3	Input validation, data preprocessing, Flask integration
3	Database / Storage	1	Crop recommendation dataset (CSV) and trained .pkl model
4	API / Integration	1	Flask integration between frontend and machine learning model
5	Security / Authentication	0	Not implemented in the current version